Varshad Potle

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Education

Government College of Engineering, Amravati **2023 – 2027**
Bachelor of Technology in Computer Science *CGPA: 8.74*

M.H.J College of Science and Commerce, Nagpur **2022 – 2023**
12th Grade *72%*

Carmel Academy, Nagpur **2020 – 2021**
10th Grade *93%*

Technical Skills

Languages: C++, JavaScript, TypeScript, SQL
Backend: Drogon (C++), Node.js, Express.js, Redis, PostgreSQL
Frontend: React.js, Next.js, Tailwind CSS, Shadcn UI
Systems & Observability: Linux Internals (/proc), Prometheus, Grafana, Docker
Developer Tools: Git, GitHub, Postman, Vercel, CMake, Ninja

Projects

GitHub Code Analyzer | *Node.js, TypeScript, LangChain, Gemini AI, Pinecone, MongoDB* **Source Code** | **Live Demo**

RAG-based AI tool to semantically query any public GitHub repository through natural language.

- Built an end-to-end ingestion pipeline using **LangChain** + **Gemini AI** to chunk and embed codebases into **768-dimensional vectors** stored in **Pinecone**.
- Engineered a **RAG query layer** with repo-namespaced vector search, retrieving **top-K** context chunks for accurate **Gemini LLM** response generation.
- Wrote **Vitest** integration tests across **2 test modules** validating ingestion accuracy and RAG retrieval correctness.

Linux Fleet Management System | *C++, Drogon, React, PostgreSQL, Prometheus* **Source Code** | **Frontend**

Distributed system for monitoring and controlling multiple Linux nodes with real-time metrics and remote command execution.

- Collected **real-time metrics from /proc** (CPU, memory, disk, network) with **zero external dependencies**, reducing agent overhead by **90%**.
- Detected **node failures within 60 seconds** via **15-second heartbeats** with exponential backoff for automatic offline marking and recovery.
- Ensured **ACID persistence** in PostgreSQL with **ON CONFLICT UPDATE**, eliminating duplicate nodes and preventing race conditions under concurrent requests.

Experience

CircuitVerse (Open Source) **Remote**
Contributor *Dec 2025 – Feb 2026*

- Diagnosed and patched a critical UI regression in the user profile module by correcting **JavaScript event propagation**, restoring navigation for the simulator platform (PR #6496).
- Collaborated with core maintainers in a high-velocity repository **(1.1k+ stars, 250+ contributors)**, ensuring strict adherence to architectural standards and code review protocols.

Achievements

Postman Student Expert: Earned industry-recognized certification demonstrating strong **software testing fundamentals**, with advanced proficiency in API development, **test cases automation**, and collection management.

Algorithmic Proficiency: Showcased excellent **analytical thinking and problem-solving ability** by systematically solving **500+ LeetCode problems** in **C++**, mastering complex **data structures & algorithms** and optimization patterns.